11827 - Probing the Faintest Fast Radio Burst Host Galaxy with JWST

Cycle: 5, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Mohit Bhardwaj (PI)	**Indian Institute of Technology Kanpur**
Dr. Jason Hessels (CoI) (ESA Member) (CoPI)	Stichting Astronomisch Onderzoek in Nederland (ASTRON)
Dr. Gourav Khullar (CoI) (CoPI)	University of Washington
Mr. Dante Hewitt (CoI) (ESA Member) (CoPI)	Universiteit van Amsterdam
Prof. Maura A. McLaughlin (CoI) (CoPI) (US Admin CoI)	West Virginia University Research Corporation

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	imaging_FRB	NIRCam Imaging	(1) NAME-FRB-190208.J1855+46
	2	nirspec_fixed_slit	NIRSpec Fixed Slit Spectroscopy	(1) NAME-FRB-190208.J1855+46

ABSTRACT

Fast radio bursts (FRBs) are brief, energetic radio flashes originating from distant galaxies, yet the nature of their sources remains unknown. Studying their host galaxies provides vital clues about their origins, energetics, and environments. While most localized FRBs are associated with relatively bright galaxies, the recently discovered FRB 20240512A stands out: it is precisely localized to an exceptionally faint galaxy, unresolved in ground-based imaging and undetected in deep optical stacks down to r>27.3 AB mag. This makes it the faintest known FRB host candidate to date. We propose JWST NIRCam imaging and NIRSpec spectroscopy to determine the redshift and physical properties of this host. The combination of JWST's sensitivity and spatial resolution is uniquely suited to detect and characterize this faint galaxy, where even state-of-the-art ground-based facilities fall short. Confirming the redshift will anchor the FRB’s energetics and contribute a crucial new data point to the Macquart relation. Spectroscopy will further reveal whether this host is a metal-poor dwarf galaxy, a dusty compact starburst, or a quiescent low-mass system—shedding light on the diversity of FRB birth sites. This program targets the faintest possible regime of FRB hosts and demonstrates the

unparalleled role JWST can play in studying the hidden corners of the transient universe.

OBSERVING DESCRIPTION

We request JWST observations of FRB 20190208A, a repeating fast radio burst localized to 260 mas precision with the EVN. Its faint host galaxy is securely identified in GTC imaging at r = 27.3 AB mag, making it the faintest confirmed FRB host to date. Ground-based spectroscopy is not feasible.

Our goal is to measure the redshift and physical properties (SFR, stellar mass, metallicity) of the host galaxy.

We propose:

NIRSpec spectroscopy with the CLEAR/PRISM grating and S400A1 slit, using a 5-point dither and spatial subpixel pattern to improve sampling and mitigate systematics. Total exposure time: 27,057s.

NIRCam imaging in the F150W filter with a FULL box dither pattern to anchor the continuum for SED modeling and align the source for spectroscopy. Total exposure time: 1,203s.

Together, these observations will enable the first spectroscopic characterization of this extremely faint FRB host, providing key constraints on progenitor models in low-mass galaxies.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NAME-FRB-190208.J1855+46	RA: 18 54 11.2700 (283.5469583d) Dec: +46 55 21.67 (46.92269d) Equinox: J2000	Parallax: 0" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Neutron stars, Pulsars]
Extended=YES

Observation

Proposal 11827, Observation 1: imaging_FRB

Fri Mar 13 22:03:01 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
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Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Neutron stars, Pulsars]
Extended=YES

Template

Module	Subarray	Target Placement
ALL	FULL	Module A center (small extended source)

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	FULLBOX	4	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W2	F322W2	DEEP8	2	1	4	4	1202.518	276970

Special Requirements

Fiducial Point Override NRCAS_FULL

Observation

Proposal 11827, Observation 2: nirspec_fixed_slit

Fri Mar 13 22:03:01 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Data Excess over lower threshold

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

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(1)	NAME-FRB-190208.J1855+46	RA: 18 54 11.2700 (283.5469583d) Dec: +46 55 21.67 (46.92269d) Equinox: J2000	Parallax: 0" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Neutron stars, Pulsars]
Extended=YES

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	1 NAME-FRB-190208.J1855+46	WATA	FULL	CLEAR	NRSRAPID	3	1	1	42.947	276970

Template

HFF Readout Mode	Slit	Subarray
false	S400A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	5	SPATIAL

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S400A1	NRSRAPID	20	12	1	WAVECAL	10	120	27056.66	276970

Special Requirements	No Parallel Attachments